



Pharm*Achieve*

OSTEOARTHRITIS

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

AE Adverse Effect(s)

BMI Body Mass Index

CBC Complete Blood Count

CRP C-Reactive Protein

DVT Deep Vein Thrombosis

ESR Erythrocyte Sedimentation Rate

MRI Magnetic Resonance Imaging

OA Osteoarthritis

PRP Platelet-Rich Plasma

RA Rheumatoid Arthritis

PATHOPHYSIOLOGY

Osteoarthritis (OA) is a chronic progressive disease of the synovial joints

- Originally believed to be a simple degenerative process of aging cartilage, OA is now recognized as a **highly complex condition associated with biochemical and inflammatory changes** within the cartilage matrix, joint cavity, joint capsule and supporting structures
- OA results from imbalance between joint destruction and repair
- Treatment is usually only initiated at the end stages of the pathologic process when symptoms begin to appear

Primary OA

Not linked to any particular cause; related to but not caused by aging

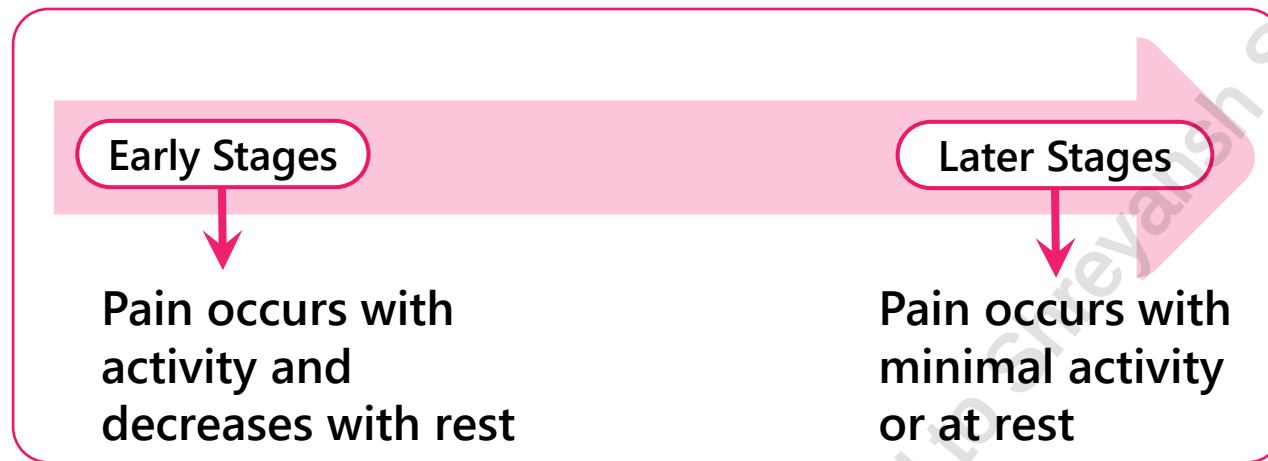
Secondary OA

Due to trauma or disease

CLINICAL PRESENTATION AND RISK FACTORS

CLINICAL PRESENTATION

Symptoms include: limited range of motion, aching pain, weakness, stiffness, joint deformities, **morning stiffness lasting < 30 minutes**; symptoms are **localized to affected joints and improve with rest**



Most affected joints: hands and fingers, knees, hips, cervical and lumbar spine, first metatarsophalangeal joint of toe



RISK FACTORS

Primary OA

- Advancing age (note: at > 65 years of age, most people will have radiographic evidence of OA but not clinical signs)
- Females
- Obesity (BMI > 30)
- Genetic susceptibility
- Congenital abnormalities

Secondary OA

- May occur in younger individuals due to previous trauma or damage
- Trauma includes sports, occupation, and repetitive joint use

INVESTIGATIONS

Drug/Disease Causes

- Intraarticular corticosteroids (joint damage)
- Nitrofurantoin (infrequent: arthralgia)
- High dose statins (joint pain)
- Bursitis, tendinitis
- Primary or metastatic bone cancer
- Conditions affecting bone (e.g. osteonecrosis, Paget's disease)
- Conditions affecting cartilage (e.g. osteochondritis dissecans)
- Rheumatoid diseases (e.g. rheumatoid arthritis, ankylosing spondylitis, pseudogout, psoriatic arthritis)

Diagnostic Evaluations

- Clinical presentation
- Symptoms
- Radiographic imaging (X-rays, MRI)
- Joint fluid analysis
- Laboratory investigations for inflammatory markers (CBC, rheumatoid factor, ESR, CRP)

RED FLAGS

- Recent history of trauma (e.g. fall, motor vehicle accident)
- Signs of injury (e.g. hematoma, edema, significant pain in affected area)
- History of osteoporosis
- Signs of DVT (painful/swollen/warm/red calf/thigh)
- Previous DVT
- Intermittent claudication (pain/cramping in calf with exertion/exercise, subsides with rest)
- Fever or other signs of infection
- Symptoms of burning, tingling, numbness (neuropathic pain)
- Morning stiffness > 30-60 minutes (consider RA)
- Symptoms improve with movement (consider RA)
- Bilateral, symmetric symptoms (consider RA)
- Patient presenting to the pharmacy with a **new onset** of joint pain or duration of pain lasting more than 10 days



GOALS OF THERAPY

- ✓ Reduce pain and improve function
- ✓ Improve quality of life, mobility and dexterity
- ✓ Minimize functional disability and improve physical functioning

Therapeutic Alternatives: Prevention

- Improves quality of life
- Physical activity and weight loss
- Avoidance of trauma and inappropriate mechanical stress on predisposed joints

NON-PHARMACOLOGICAL MEASURES

Strongly recommend

- Exercise
- Self-efficacy and self-management programs
- Knee and hip OA
 - Weight loss
 - Tai Chi
 - Cane
- Knee: Tibiofemoral knee brace
- Hand: 1st carpometacarpal orthosis



Conditionally recommend

- Heat, therapeutic cooling
- Cognitive Behavioral Therapy
- Acupuncture
- Knee and hand: kinesiotaping
- Knee and hip: Balance training
- Knee:
 - Patellofemoral knee brace
 - Yoga
 - Radiofrequency ablation
- Hand:
 - Other hand orthoses
 - Paraffin

PHARMACOLOGICAL MEASURES

Strongly recommend

- Knee, hip and hand: oral NSAIDs
- Knee and hip: intraarticular steroids
- Knee: topical NSAIDs



Minimum **2-4 weeks** to assess therapeutic response



Therapy is patient-specific! Take into consideration all patient factors.

Conditionally recommend

- Knee, hip and hand:
 - Acetaminophen
 - Duloxetine
 - Tramadol
- Hand:
 - Topical NSAIDs
 - Intra-articular steroids
 - Chondroitin
- Knee: Topical capsaicin

RECOMMENDATIONS AGAINST

Conditionally recommended AGAINST for knee, hip or hand OA

- Non-tramadol opioids
- Colchicine
- Fish oil
- Vitamin D
- Knee and hip: intra-articular botulinum toxin injections
- Knee and hand: intra-articular hyaluronic acid
- Hand: topical capsaicin

Strongly recommended AGAINST for knee, hip or hand OA

- Bisphosphonates
- Glucosamine
- Hydroxychloroquine
- Methotrexate
- TNFI and interleukin-1 receptor antagonists
- Knee and hip:
 - Chondroitin
 - PRP treatment
 - Stem cell injections
- Hip: intra-articular hyaluronic acid

NSAIDS SELECTION

		GI RISK 		
		LOW	MEDIUM	HIGH
CV RISK 	LOW	Low-dose nonselective NSAID	Low-dose nonselective NSAID + Gastroprotection or Low-dose celecoxib	Low-dose celecoxib + Gastroprotection
	MEDIUM	Low-dose naproxen	Low dose naproxen + Gastroprotection or Low-dose celecoxib	Low-dose celecoxib + Gastroprotection
	HIGH	Consider alternative therapy (e.g. duloxetine, local injections) or Low-dose naproxen	Consider alternative therapy (e.g. duloxetine, local injections) or Low-dose naproxen + Gastroprotection	Consider alternative therapy (e.g. duloxetine, local injections)

PHARMACOLOGICAL MEASURES

DRUG	COMMENTS
<i>TOPICAL NSAIDs</i> (e.g. diclofenac)	• Strongly recommended for knee/conditionally recommended for hand (small joints); commonly used in elderly • Should be considered prior to oral NSAID therapy as topicals are considered 1st line • No recommendation for hip OA • Safe to use with oral therapies • Faster onset than capsaicin (onset 1-2 weeks)
<i>ORAL NSAIDs</i> (including COX-2 inhibitors)	• Strongly recommended for hip, knee, and hand OA • Mainstay of oral pharmacological management and use is strongly recommended • Large number of trials have demonstrated their short-term efficacy • Doses as low as possible and continued for as short a time as possible • Always weigh risk versus benefit due to risk of CV, GI and renal side effects
<i>ORAL ACETAMINOPHEN</i>	• Conditionally recommended; meta-analysis suggest monotherapy may be ineffective (long term treatment no better than placebo for most individuals) • Use in those with limited options due to intolerance or CI to NSAIDs • May increase anticoagulant effect of warfarin (dose dependant) • Increased risk of hepatotoxicity with concurrent alcohol use
<i>TOPICAL CAPSAICIN</i>	• Conditionally recommended in knee OA - limited evidence • Conditionally recommended AGAINST in hand OA • Safe to use with oral therapies or as alternative if oral therapy is not tolerated • Onset 2-4 weeks; transient burning and stinging (not well tolerated- not used much)

PHARMACOLOGICAL MEASURES

FYI

DRUG	COMMENTS
ORAL OPIOIDS	• Tramadol conditionally recommended for patients with knee, hip or hand OA (i.e. if an opioid is being considered, tramadol is conditionally recommended) • Non-tramadol opioids conditionally recommended against unless alternatives have been exhausted • Multiple AEs - not considered to be first line agents; individualize therapy to each patient
ORAL DULOXETINE	• Conditionally recommended for patients with knee, hip or hand OA • May be used for patients with concurrent depression and/or neuropathic pain • Risk of serotonin syndrome when combined with other serotonergic drugs
INTRA-ARTICULAR GLUCOCORTICOIDS	• Strongly recommended for patients with hip or knee OA, conditionally for hand OA • Conditionally recommended over other forms of intra-articular injections for knee and hip OA • Pain relief within 24-48h, duration weeks-months
ORAL GLUCOSAMINE, CHONDROITIN	• Chondroitin conditionally recommended for hand OA – limited evidence • Glucosamine strongly recommended AGAINST for knee, hip or hand OA • Chondroitin strongly recommended AGAINST for knee and hip OA • Glucosamine toxicity potential is low, although some patients may show elevations in serum glucose levels

MONITORING PARAMETERS

- ✔ Patient's pain level (digital or numeric scale can be used)
 - ✔ First thing in the morning
 - ✔ After exertion
- ✔ Functional status (performance of basic everyday tasks)
- ✔ Quality of life
- ✔ Drug adverse effects



Once stabilized on a drug regimen, the patient should be re-assessed between 2-4 weeks initially to plan for next steps to reach optimal function

TAKEAWAY

- OA is a chronic, progressive disease of synovial joints
- Morning stiffness typically lasts < 30 minutes and symptoms improve with rest
- **Strongly recommended:**
 - Oral NSAIDs
 - Knee: topical NSAIDs
 - Knee and hip: intraarticular steroids
- Long-term acetaminophen monotherapy similar to placebo; benefit unlikely
- **Recommend against:** colchicine, fish oil, glucosamine

CASE 1

AB is a 68-year-old female (height: 5'2 and 230 lbs) who complains of her left knee feeling quite stiff for the first 15-20 minutes in the morning when she wakes up. She says her pain is usually worse after physical exertion but improves with rest. She has a past medical history significant for hypertension, depression, dyslipidemia, and headaches. She takes perindopril/indapamide 4/1.25 mg po daily, sertraline 100mg PO once daily, atorvastatin 10mg PO QHS, ibuprofen 200mg PO PRN for headaches, and Aspirin® 81mg PO once daily. She smokes 1 pack per day and has environmental allergies.

1. Which of the following statements regarding osteoarthritis is NOT correct?
 - a) Joint stiffness normally lasts > 30 minutes in the AM
 - b) OA is recognized as a condition with biochemical and inflammatory changes
 - c) Joint stiffness normally lasts < 30 minutes in the AM
 - d) Patients with OA mostly suffer from pain and joint function impairment



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2. Which of the following is a modifiable risk factor that AB has for osteoarthritis?

- a) Hypertension
- b) Obesity
- c) Headaches
- d) Depression



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3. Which of the following antidepressants may be considered in patients with comorbid osteoarthritis?

- a) Citalopram
- b) Duloxetine
- c) Paroxetine
- d) Fluoxetine



CASE 1

AB is a 68-year-old female (height: 5'2 and 230 lbs) who complains of her left knee feeling quite stiff for the first 15-20 minutes in the morning when she wakes up. She says her pain is usually worse after physical exertion and improves with rest. She is seeking advice. She has a past medical history significant for hypertension, depression, dyslipidemia, and headaches. She takes amlodipine 10 mg PO QHS, sertraline 100 mg PO once daily, atorvastatin 10 mg PO QHS, ibuprofen 200 mg PO PRN for headaches, and ASA 81 mg PO once daily. She smokes 1 pack per day and has environmental allergies.

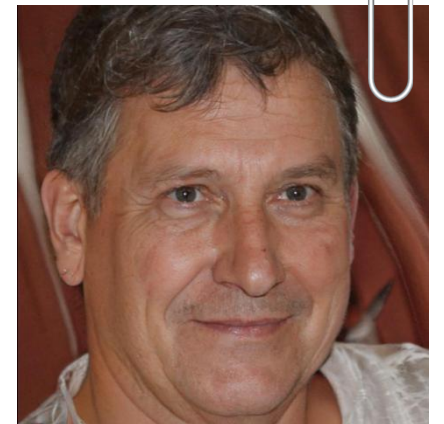
4. AB asks for your advice on pain management. Which of the following would NOT be an appropriate recommendation?
- a) Naproxen 250-500 mg PO BID
 - b) Voltaren Emulgel® (diclofenac) 2.32% applied to affected joint two times a day as needed
 - c) Refer to the physician to discuss intra-articular steroid injections
 - d) Glucosamine-chondroitin supplement



CASE 2

PJ is a 60-year-old male (height: 5'10 and 180 lbs) who complains of left hip pain. He been taking acetaminophen 1000 mg po QID for years but feels this is no longer working. PJ states that he also uses topical diclofenac gel occasionally. He has a past medical history of anxiety, dyslipidemia, hypertension and occasional headaches. He takes escitalopram 10 mg PO daily, rosuvastatin 40 mg PO daily, ramipril 10 mg PO daily, and amlodipine 5 mg PO daily. He also takes a vitamin E supplement once daily and calcium carbonate 500 mg PO BID. He is a social drinker, and he is allergic to shrimp and pollen.

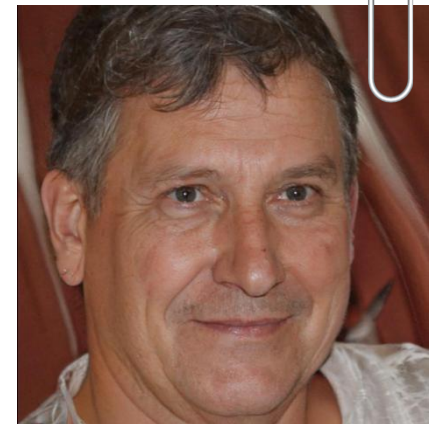
1. Which of the following would be the most appropriate recommendation for PJ?
 - a) Suggest PJ see his physician to trial 2 weeks of tramadol
 - b) Suggest he use the topical diclofenac on a more regular basis
 - c) Suggest a 2–3-week trial of an oral NSAID, such as naproxen
 - d) Suggest the addition of OTC chondroitin monotherapy



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2. Which of the following would NOT be an appropriate non-pharmacological therapy to recommend to PJ?
- a) Balance exercises
 - b) Thermal interventions (i.e. application of heat/cold as needed)
 - c) Massage therapy
 - d) Assistive device (e.g. cane, walker)

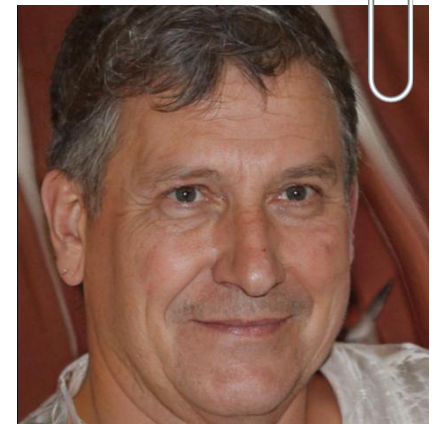


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3. When should you follow up with PJ to assess whether recommended therapy is appropriate?

- a) 1-2 days
- b) 2-4 weeks
- c) 1-2 weeks
- d) 2-4 days

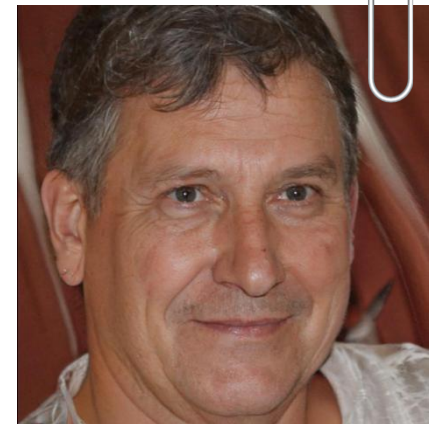


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4. Which of PJ's medications have the potential to cause symptoms which may mimic osteoarthritis?

- a) Rosuvastatin
- b) Ramipril
- c) Amlodipine
- d) Escitalopram



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CHANGE LOG

May 2025

- Slide 14: Added that intraarticular glucocorticoid injections are strongly recommended for hip or knee OA based on guidelines

Aug 2025

- Slide 5: minor wording change
- Slide 8: Changed to “lasting >30-60 minutes”
- Slide 14: Added onset for intra-articular glucocorticoids

Nov 2025

- Formatting changes made
- Slide 3: Updated legend
- Slide 6: Minor phrasing changes
- Slide 12: Relocated. Added non-tramadol opioids
- Slide 14: Added ‘limited evidence’ to capsaicin
- Slide 15: Added ‘limited evidence’ for chondroitin
- Slides 18-25: Minor wording, formatting and grammar changes

April 2026

- Slide 13: Minor update to capsaicin
- Slide 14: Added conditionally for hand under IA glucocorticoids
- Slide 20: Minor update to option B
- Minor wording changes throughout